AMAZON Q DEVELOPER (LEC: - 26)

Amazon Q Developer: -

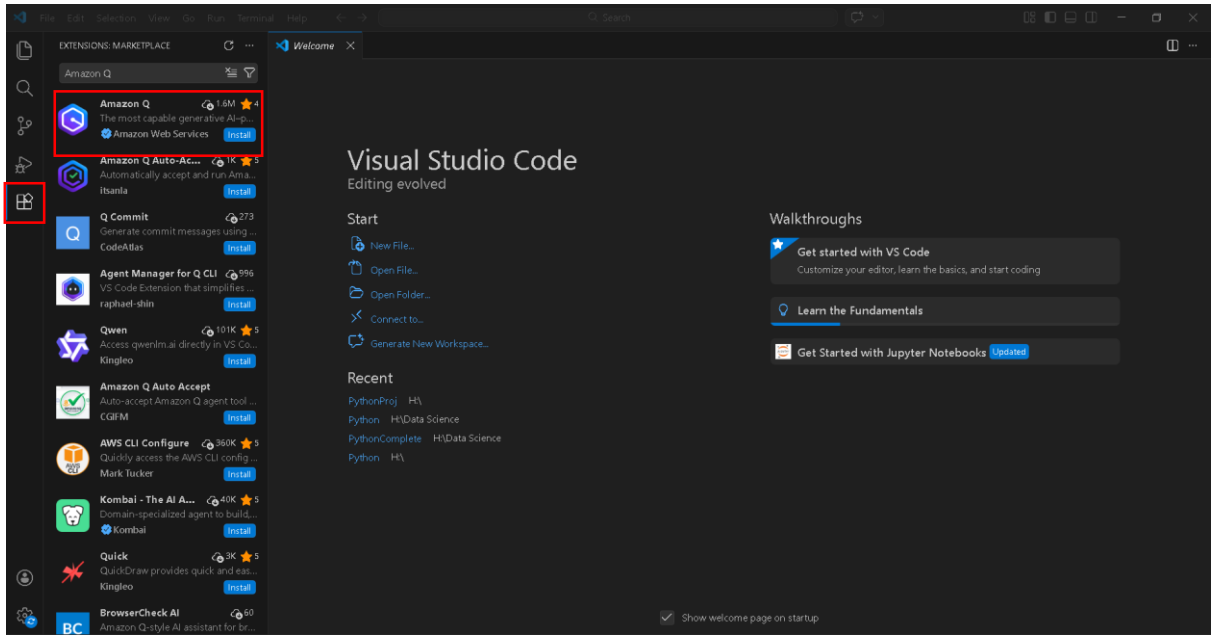
- Amazon Q Developer is an AI-powered assistant that helps developers build, analyze, and manage applications on AWS by generating code, answering questions, and improving software quality.
- It can Assist in coding and testing.
- It supports all the major programming language like java, python, Go etc.
- It helps in diagnosing errors
- It can also perform security scanning and fixes.
- Where to Access Amazon Q – Developer from: -
    - AWS Management Console
    - AWS CLI
    - IDE such as VS code

How Amazon Q can helps in Software Development Lifecycle: -

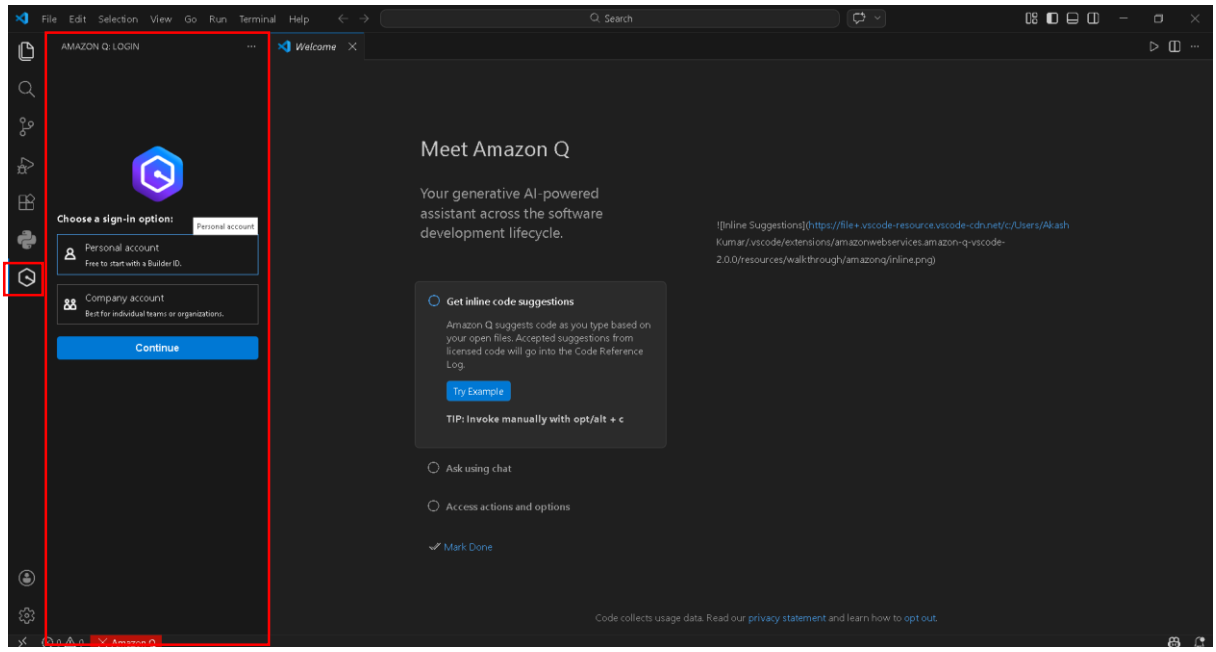
- **Planning: -**
    - Business Requirement Document(Epic) : -
        - Ex: - A store inventory manager, I should be able to check the inventory details for items in the store and update them when new items are received at the store.
    - Features and User Stories: -
        - Based on Epic we create our Features and User Stories.

    - For this phase we can use Amazon Q Business
    - In Amazon Q Business Jira Plugin and you can write the requirements in the Natural language in the chat interface.
    - And then it creates Features and User stories automatically in the JIRA.

- **Design: -**
    - In the Design Phase Amazon Q Developer can help you generate Architecture for the use case.
    - So now Let's see how: -
        - The first thing you need to download VS Code.



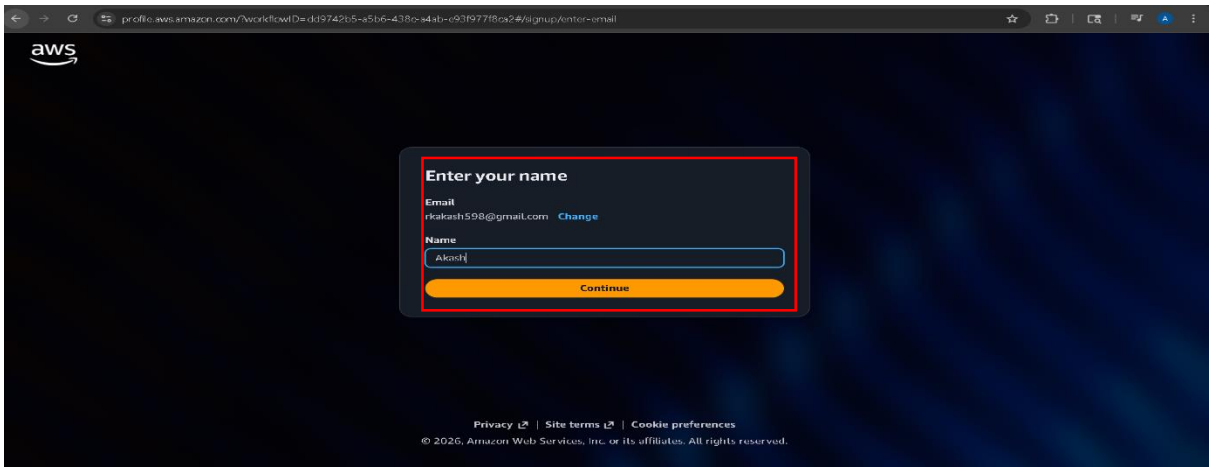
- Open it and click on Extension icon (left side Fifth icon)
- And Search Amazon Q
- Click on it and install it



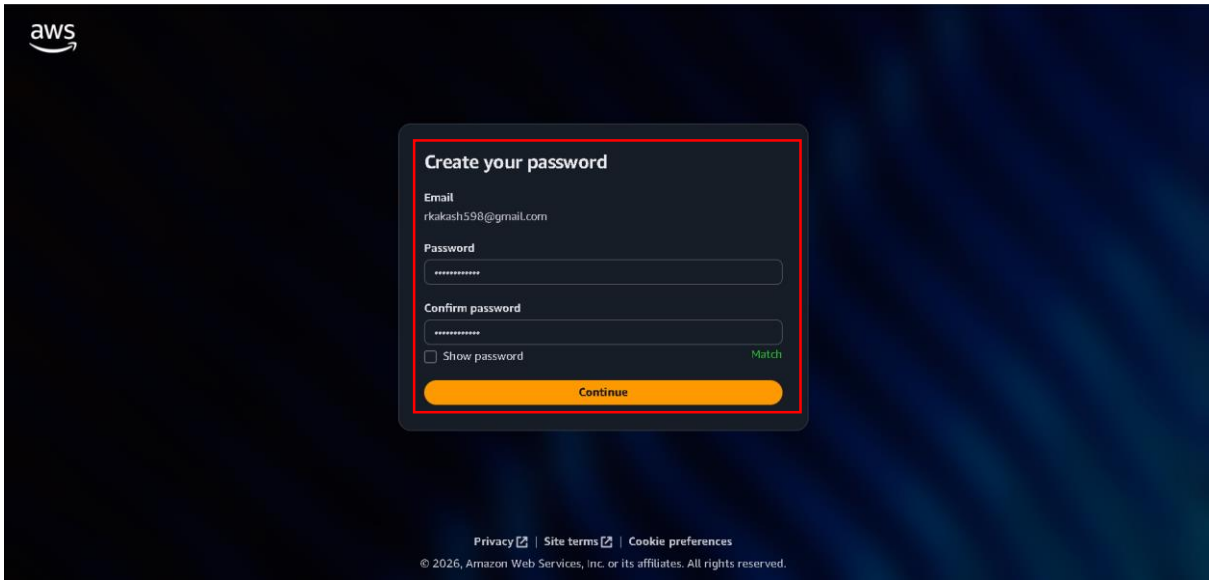
- Click on sign in with personal account.
- But for sign in you need a Builder id

So let's create a Builder id: -

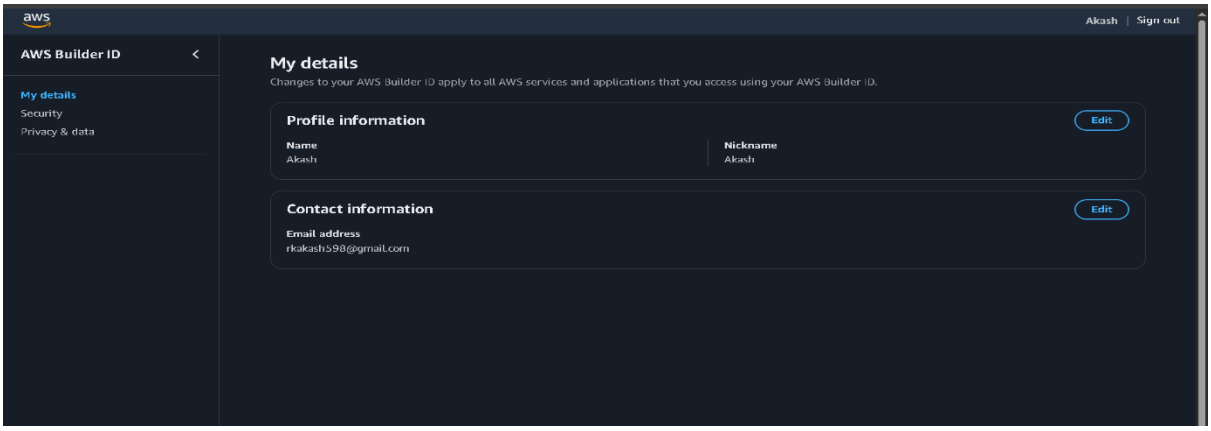
- Open any Browser and search AWS Builder ID.



- Enter Email and Name and click continue
- Enter verification code

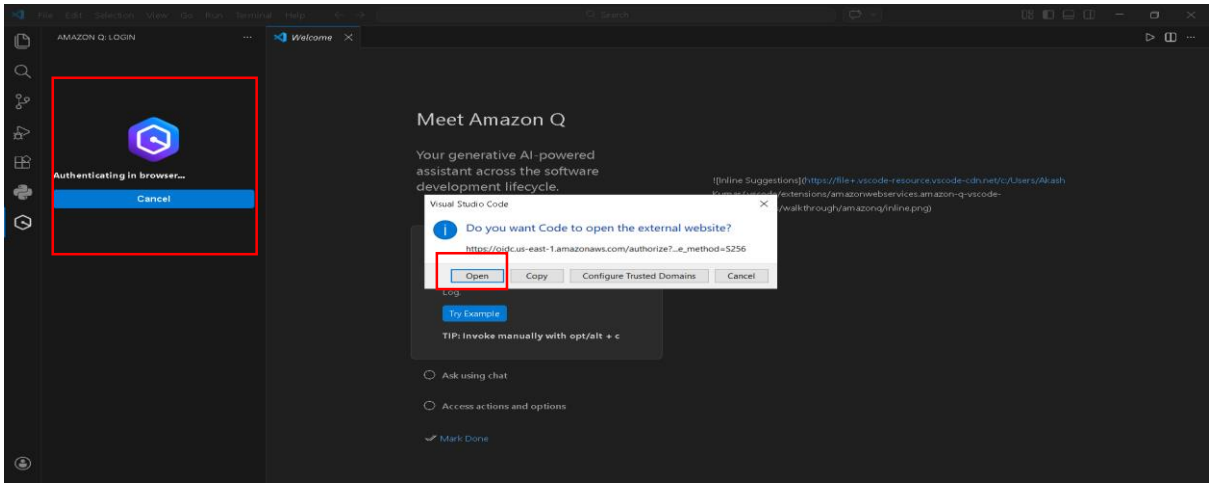


- Now Enter Password

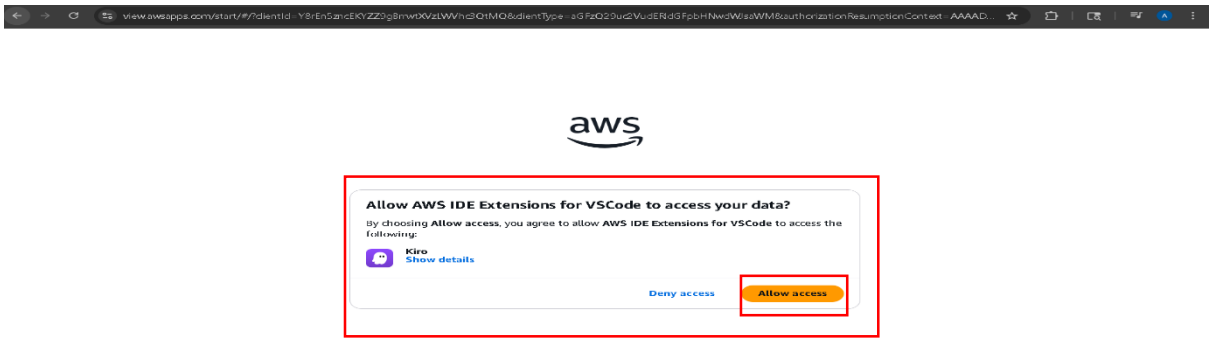


- Now your Builder id is Ready

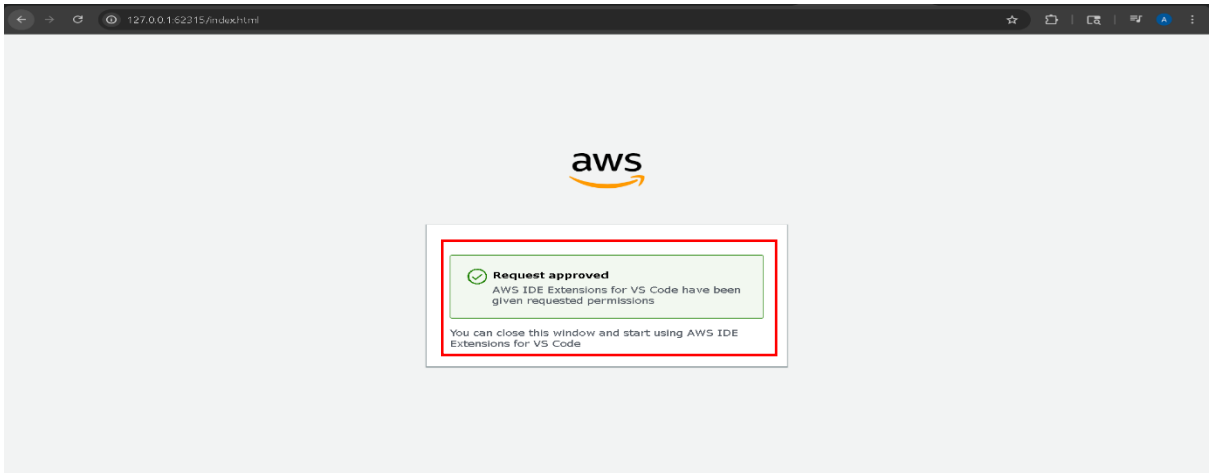
- Now Back to your Vs Code and go in Amazon Q
- Choose Sign in personal account
- Click continue.



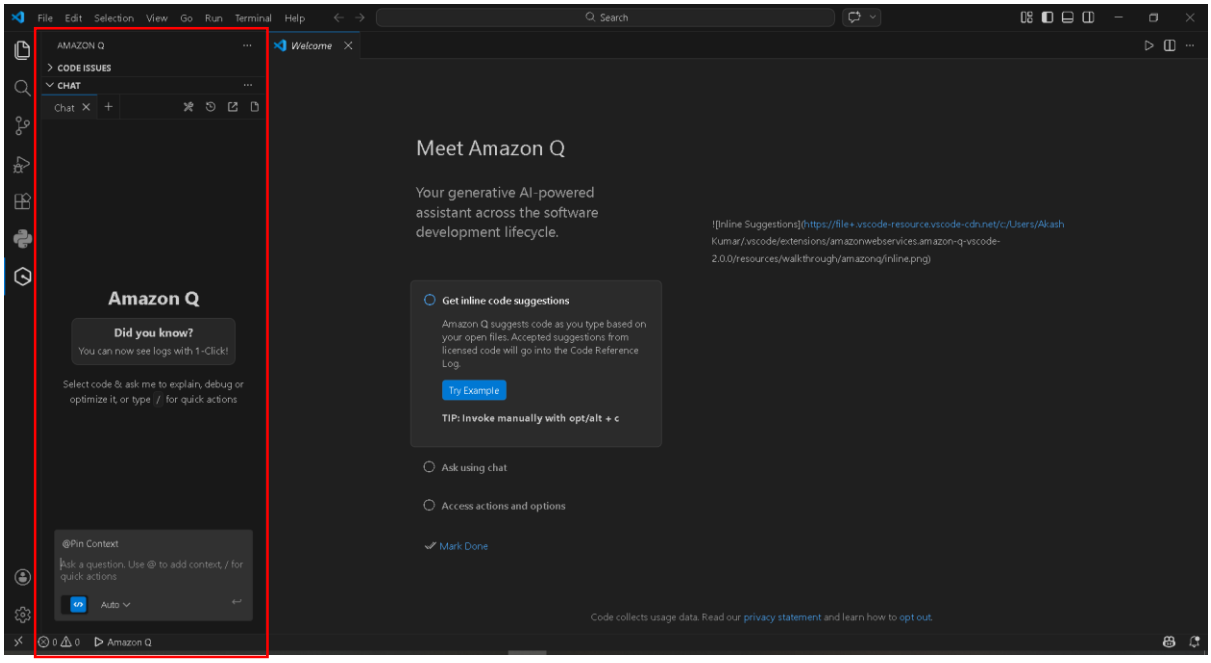
- Now click on Open
- Redirected to your browser



- Now Click Allow Access

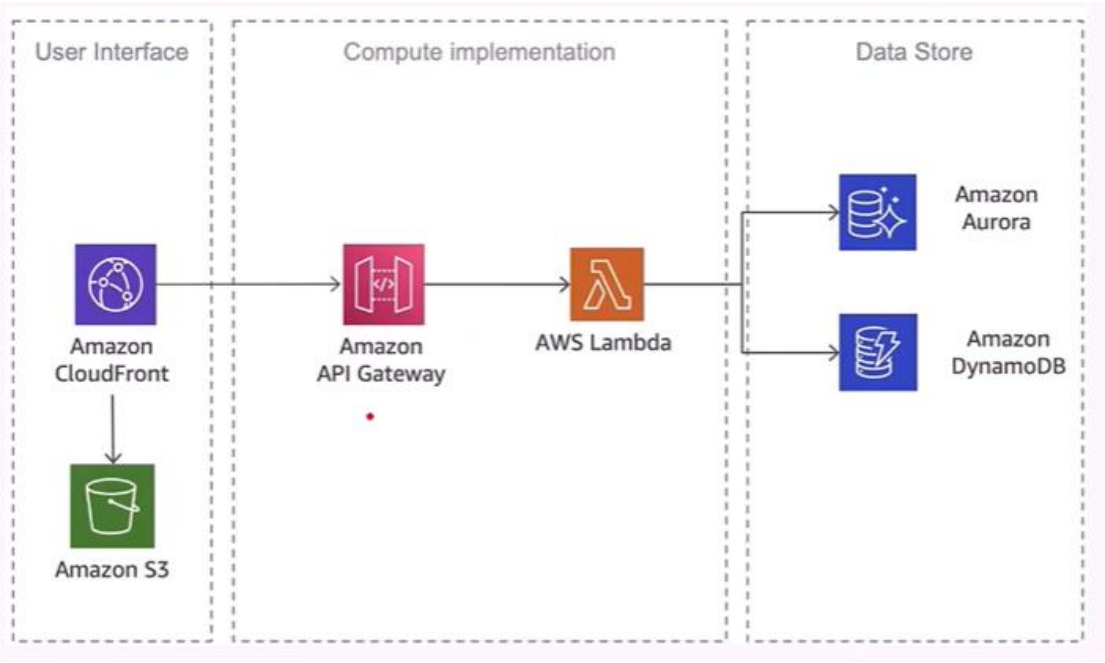


- Your request is approved



- Now our Amazon Q id ready ask any Question it's a LLM

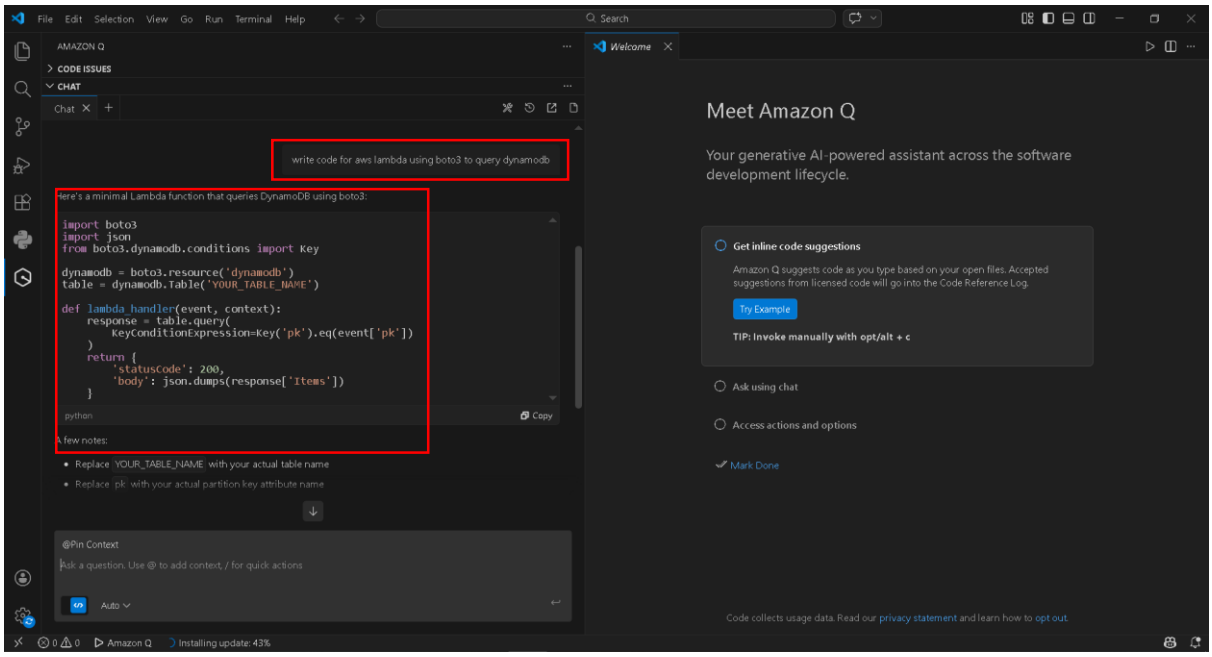
Now Understand about AWS Micro-Services Architecture: -



- **Now this is our Flow: -**

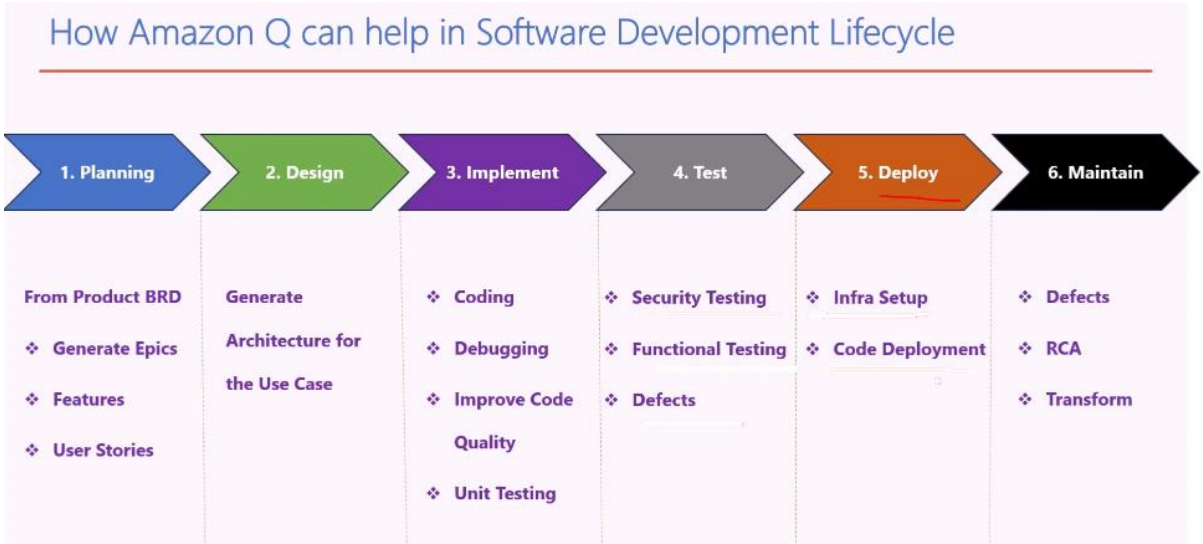
  - We will Send the request to Amazon API gateway
  - Which invokes the Amazon Lambda Function
  - And then Amazon Lambda makes a query to the Amazon Dynamo DB

- **Implementation: -**
  - Coding
  - Debugging
  - Improve code
  - Unit Testing
- **Example : -Now let's generate the code for this Architecture: -**
  - Now let's we start coding
  - In which we'll have write the Lambda function which will query the Amazon Dynamo DB
  - And we are going to use python SDK Boto3 for doing this.
  - So let's Ask Amazon Q Developer to Generate the code



  - Now you see it write code for us.
  - Also explain and fix your already written code.
- **Testing: -**
  - Security Testing
  - Functional Testing
  - Defects
- **Deploy: -**
  - Infra setup
  - Code Deployment

- **Maintain: -**
    - Defect
    - RCA
    - Transform



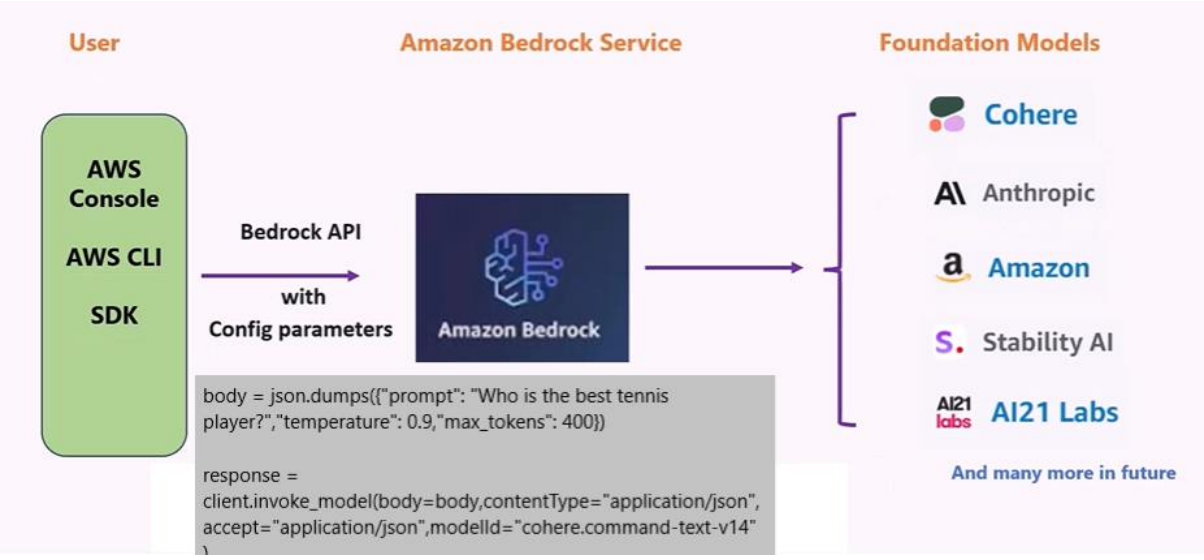
NOTE: - This Lecture to give you aa view of how Amazon Q Developer can help across the software development.

Amazon Bedrock (Lec: - 28)

Amazon Bedrock: -

Amazon Bedrock is a cloud service that provides access to foundation AI models (like text, image, and chat models) via APIs to build generative AI applications easily.

How does Amazon Bedrock work – High Level: -



- So you can think amazon Bedrock service as an interface which allow you to access the foundation model that we see in the right side.
- You are a user and you want to access foundation model
- So let's see how we can access any foundation model through Amazon Bedrock.
    - So what you need to do is just need to make an API Request with some configuration parameter to Amazon Bedrock service.
    - Ex: prompt: -who is the best Tennis player?

      ```
      body = json.dumps({"prompt": "Who is the best tennis player?",
      "temperature": 0.9, "max_tokens": 400})

      response = client.invoke_model(body=body, contentType="application/json",
      accept="application/json",modelId="cohere.command-text-v14")
      ```

    - All things is taken care by Amazon Bedrock.
    - It is pay as you go
    - So if you make Ten API request , you will be charged based on thatIf you
    - don't make any API request for Five day's you will not be charged